CMA Study Notes

Cost & Management Accounting (EPGP, Section C)

Day 1 to Day 4 lecture notes + Problem Sets 1, 2 and 3 + Excel step-by-step

How to use these notes

Each topic has a **yellow box** (plain-language explanation), a **blue box** (the professor's method, step by step), a **green box** (how to do it in Excel) and a **red box** (exam tips / traps). Read the yellow box first, then the problem walkthroughs.

Sources covered

Lecture decks: Day 1 (Intro to Management Accounting), Day 2 (Cost terms), Day 3 (Manufacturing costs and cost flow), Day 4 (Overheads, allocation of service departments, common costs). Problem Set 1 (cost sheets) with solutions; Problem Set 2 (inventory back-solving); Problem Set 3 (overhead distribution) with the professor's Excel workbook; PlayCraft question with solution.

Note on corrections

A few typing slips in the source solutions were corrected and are flagged in the notes: Set 1 P1 cost of production (706,625 not 706,620); Set 3 P2 Stores split (600 and 400); PlayCraft part (b) direct material consumed (270,000 not 200,000). In Set 3 P4, the solution is not in the files, so it was worked from the question and checked (totals tie to 26,840).

Contents

PART A: Day 1: Introduction to Management Accounting 4

1. What accounting systems do . . . 4
2. Management accounting vs Financial accounting . . . 4
3. Cost accounting vs Management accounting . . . 5
4. Strategy and management accounting . . . 5
5. What is managerial accounting? . . . 5
6. Value, customer value propositions and the value chain . . . 5
7. The five-step decision-making process (planning & control) . . . 6
8. Three guidelines for management accountants . . . 6
9. Professional ethics (IMA standards) . . . 6
10. Value-added vs non-value-added activities . . . 7

Class participation answers (CP) . . . 7

PART B: Day 2: Cost Terms, Classifications and Purposes 8

1. Basic cost terminology . . . 8
2. Direct and indirect costs . . . 8
3. Cost behaviour: variable, fixed, mixed . . . 8
4. Multiple classifications (BMW X6 style) . . . 9
5. Opportunity cost, sunk cost, relevance . . . 9
6. Answers to the class-participation questions . . . 10
7. Types of firms and inventories . . . 10

PART C: Day 3: Manufacturing Costs and the Flow of Costs 11

1. The three manufacturing (product / inventoriable) costs . . . 11
2. Non-manufacturing costs (period costs) . . . 11
3. Professor's "factory funnel" method: how costs flow . . . 12
4. Income statement layout (professor's order) . . . 13
5. Now You Do It: Solomon Grundy Corp. . . . 13
6. Answer to Question for Thought 5 (Madi Corporation) . . . 14

PART D: Day 4: Overhead Accounting, Analysis and Control 15

1. Key definitions . . . 15
2. Production overhead: five steps . . . 15
3. Secondary distribution: bases for service departments . . . 16
4. Single-rate vs dual-rate allocation . . . 16
5. Three methods for allocating service department costs . . . 16
6. Administration overhead and selling & distribution overhead . . . 17
7. Allocating common costs . . . 17

PART E: Problem Sets: Solutions in the Professor's Method, with Excel Steps 18

The master skeleton (use for every problem) . . . 18

Excel basics used in all problems . . . 18

SET 1: Cost sheets (Problem Set 1) . . . 19
SET 2: Back-solving inventories and absorption costing (Problem Set 2) . . . 24
SET 3: Overhead distribution (Problem Set 3) . . . 27
Practice problem: PlayCraft Inc. (toys): T-accounts, COGM, COGS, income statement . . . 32
PART F: Quick Revision Sheet **35**
One-page formula sheet . . . 35
Common mistakes to avoid . . . 35
Glossary (A to Z) . . . 35

PART A: Day 1: Introduction to Management Accounting

In simple words (layman view)

Accounting is the **language of business**. Managers use numbers the way a pilot uses a cockpit dashboard: to see where the company is, decide where to go, and check whether it is getting there. Management accounting is the part of accounting built **for the people inside the company** so they can make better decisions.

1. What accounting systems do

- Accounting systems turn economic events and transactions into information managers can use. Data is **collected, categorised, summarised and analysed**.
- They feed two kinds of output: the **financial statements** (for outsiders) and **internal performance reports** (for managers).
- Different managers need different views of the same data. Example (sales data): the sales manager wants it by region or salesperson; the distribution manager wants orders by location and due date; the manufacturing manager wants quantities ordered so production can be scheduled.
- The cost accounting system accumulates both financial and managerial accounting data.

2. Management accounting vs Financial accounting

In simple words (layman view)

Financial accounting = a **report card for outsiders** (owners, banks, government): rule-bound, historical, whole company. Management accounting = a **planning tool for insiders**: flexible, future-focused, detailed, no fixed rules. Its key idea is the **decision-making focus**, and every piece of information must pass a cost-benefit test.

Basis	Financial accounting	Management accounting
Period	After a stated period (quarter/year)	Frequent intervals, whenever management needs it
Objective	Periodic reports to owners, creditors, government	Assist internal management
Time focus	Historical data	Historical, current and **future** data (forward-looking)
Unit of expression	Monetary only	Any unit: money or statistical (kg, hours, units)
Data & sources	Actual historical, internal sources	Future plans, projections + past data; internal and external sources
Subject / coverage	Business as a whole, aggregate	Limited coverage but detailed analysis (divisions, products)
Standards	Fixed by outside parties, objective, money consequences	Fixed by management itself, subjective
Legal compulsion	Statutory, compulsory	Voluntary, optional
Precision	"Pie to pie" accuracy	May involve guesswork/estimates
Unit of account	Whole business	Divisions, units, results of segments
Publication & audit	Essential	Not required
Principles	Ind AS, GAAP, IFRS	No such principles: flexible and tailored

3. Cost accounting vs Management accounting

Basis	Cost accounting	Management accounting
Object	Determine the cost of a product	Provide information to management
Deals with	Cost data	Both cost and revenue data
Scope	Limited	Much wider
Nature	Past and present facts	Projection of future plans
Principles	Procedures exist	No fixed procedures
Type of data	Quantitative aspects	Quantitative **and** qualitative
Users	Internal and external	Internal (management)
Value added to decisions	Low	High

4. Strategy and management accounting

- **Strategy** = how an organisation matches its own capabilities to opportunities in the market. Two broad choices: **cost leadership** (good product at a low price) or **product differentiation** (unique product at a higher price).
- **Strategic cost management** = cost management that focuses on strategic issues.
- Management accountants help by showing the sources of competitive advantage: (1) our cost/productivity advantage over competitors, or (2) the premium price we can charge compared with the cost of adding distinctive features.

Answer to Question for Thought 1

Question for thought 1: can a firm follow a strategy "in the middle"? Usually this is risky ("stuck in the middle"): you are neither the cheapest nor the most unique, so customers have no clear reason to pick you. It can work only if the firm has real scale or technology that lets it be reasonably cheap *and* reasonably differentiated at the same time. Most successful firms still choose one primary position and support it with the other.

5. What is managerial accounting?

Managerial accounting is the process of **Identifying, Measuring, Analysing, Interpreting and Communicating** information in pursuit of the organisation's goals and strategy. It is an integral part of the management process and provides information critical to **planning, control and decisions**.

How it adds value (objectives):

- Provide information for decision-making and planning.
- Help managers direct and control activities.
- Motivate managers and employees toward the organisation's goals.
- Measure performance of sub-units, activities, managers and employees.
- Assess the organisation's competitive position.

6. Value, customer value propositions and the value chain

- **Value** = the usefulness a customer gets from a product or service. The whole customer experience (quality, timely delivery, fair price) decides value.
- **Customer value propositions** (three strategies): **Operational excellence** (faster, more convenient, cheaper), **Product leadership** (higher-quality products), **Customer intimacy** (understand and respond to individual customer needs).

The value chain is the sequence of business functions in which a product is made progressively more useful to customers:

Step	What happens
1. Research & development	Generating and experimenting with ideas for new products, services, processes

Step	What happens
2. Design of products & processes	Detailed planning, engineering and testing
3. Production	Procuring, transporting, storing, coordinating and assembling resources to make the product / deliver the service
4. Marketing	Promoting and selling
5. Distribution	Processing orders and shipping to customers
6. Customer service	After-sales service
Administration (supports all six)	Accounting, HR, IT

Key success factors customers expect from every link: **Cost & efficiency, Quality, Time** (new-product development time and customer-response time), **Innovation**, and **Sustainability** (long-term financial, social and environmental goals).

7. The five-step decision-making process (planning & control)

1. Identify the problem and uncertainties.
2. Obtain information.
3. Make predictions about the future.
4. Make decisions by choosing between alternatives.
5. Implement the decision, evaluate performance and learn.

> **In simple words (layman view)**
>
> Steps 1 to 4 = **Planning** (deciding what to do). Step 5 = **Control** (doing it, checking how it went, and learning). Think of a road trip: plan the route (1 to 4), then drive and watch the dashboard and correct course (5). The **budget** is the most important planning tool: a numbers version of the plan that helps coordinate everyone.

8. Three guidelines for management accountants

- **Cost-benefit approach:** benefits of an action must exceed its costs.
- **Behavioural and technical considerations:** people are involved in decisions, not just rupees.
- **Different costs for different purposes:** use different ways of computing cost for different decisions.

9. Professional ethics (IMA standards)

Four standards: **Competence, Confidentiality, Integrity, Objectivity/Credibility** (the slide lists Objectivity; IMA text uses Credibility).

Case	What happened	Analysis
A: Leston	CFO John Young learned a competitor was likely to go bankrupt but did not tell the board while it was deciding on a plant closure.	Relevant information was withheld from decision-makers: breach of **integrity/credibility** (and objectivity). Unless the info was truly confidential, he should have disclosed it.
B: QBX	Purchasing manager bought cheaper, lower-grade Felstar. Usage rose but quality was unaffected and overall cost fell.	No ethics breach. A cost-benefit decision made in the company's interest, handled competently. (Good practice: tell production and record the reason.)
C: Central Distributing	Hillary Baxter deliberately submitted very low sales targets so she would always "look good".	**Budgetary slack**: a breach of **integrity and objectivity**; she misleads management for personal benefit.

Darius Enterprises case (Question for thought 3). Franklin split a Rs/$250,000 remodelling into $190,000 and $60,000 to avoid the $200,000 board-approval limit.

- **Effect on financial statements?** None in total: the same $250,000 of assets and cash flows are recorded either way. The effect is on **control**, not on the numbers.
- **Ethical?** No. He knowingly bypassed an internal control to avoid board approval, in a period of tight cash.

- **Standards breached:** Integrity (circumventing policy) and Credibility/Objectivity (not presenting the full picture to the board); also Competence (not following company procedures).

10. Value-added vs non-value-added activities

In simple words (layman view)

A **value-added (VA)** activity is one the customer would happily pay for. A **non-value-added (NVA)** activity only adds cost: waiting, re-doing, moving things around, fixing mistakes. Managers try to cut NVA costs without hurting quality. Classification can be **subjective** (as the professor noted).

Activity (Falcon lawn mowers)	Class	Why
1. Attaching engine to the mower body	VA	Needed to build the product
2. New air-conditioning in executive offices	NVA	Does not change what the customer receives
3. Replacing a defective wheel	NVA	Rework: fixing an earlier mistake
4. Designing/printing owner's manual for new model	VA	Customer expects and values it
5. Moving mowers to finished-goods warehouse	NVA	Handling/storage adds no value
6. Attaching handle (slow due to worker slowdown)	Both	Attaching = VA; the extra time lost to the slowdown = NVA

Company example	Class
Microsoft: coding a new spreadsheet package	VA
General Mills: painting maintenance supervisor's office	NVA
Mayo Clinic: examining a new patient	VA
American Airlines: 90 minutes a 757 sits idle between flights	NVA
Office Depot: moving paper cases within the same warehouse	NVA
Rolex: attaching a band to the watch face	VA
Postal service: reprocessing wrongly sorted mail (machine fault)	NVA
IndiaBulls: correcting errors made by staff in customer accounts	NVA
Marriott: upgrading bedding in competitive markets	VA (customers value it)

Typical NVA costs in services (from the slide):

Industry	Examples of non-value-added costs
Airlines	Vouchers given for poor service; tracing/replacing lost luggage; extra crew pay due to delays from preventable maintenance problems
Banks	Correcting bank errors; manual work during computer downtime; employee theft; defaulted loans to poor-risk borrowers
Hotels	Broken dishes/lost linen; replacing lost room keys; over-staffing the front desk off-peak; excess food costs

Class participation answers (CP)

- **Q1:** Management accounting _____. **Answer A**: it focuses on estimating future revenues, costs and other measures to forecast activities and results.
- **Q2:** Managers use management accounting information to _____. **Answer B**: communicate, develop and implement strategies.

PART B: Day 2: Cost Terms, Classifications and Purposes

In simple words (layman view)

A **cost** is what you give up to get something. Cost accounting is simply careful bookkeeping of "what did it cost us to make/do this thing, and how does that cost behave if we make more or less?"

1. Basic cost terminology

Term	Meaning	Example
Cost	A sacrificed or forgone resource to achieve a specific objective (CAS-1: measurement in money of resources used for production of goods/services)	Cost of making an appliance = materials + labour + ...
Actual cost	A cost that has already occurred	Rs 50,000 actually paid for steel
Budgeted cost	A predicted (future) cost	Rs 55,000 expected next month
Cost object	Anything for which a separate cost measurement is wanted	Product (BMW X6), service (dealer hotline), project (R&D), customer, activity (machine set-up), department

- **Cost accumulation** = collecting cost data in an organised way through the accounting system.
- **Cost assignment** = attaching accumulated costs to a cost object, in two ways: **tracing** (costs with a direct relationship) and **allocating** (costs with an indirect relationship).

In simple words (layman view)

Cost vs expense. A cost is any sacrifice and can sit on the balance sheet as an **asset** (e.g. raw materials inventory) until used. It becomes an **expense** when its benefit is used up in the period (e.g. travel expense, a training course with no quantifiable future benefit).

2. Direct and indirect costs

- **Direct costs** can be traced to the cost object conveniently and economically (steel and tyres for a car, assembly-line wages).
- **Indirect costs** cannot be traced conveniently, so they are **allocated** in a rational, systematic way (electricity, rent, property tax, plant administration).
- Total cost of object = direct costs (traced) + indirect costs (allocated). Managers trust direct costs more.

Factors that decide direct vs indirect: (1) materiality (tiny costs are not worth tracing), (2) available information-gathering technology (more costs become direct), (3) design of operations (a dedicated plant makes more costs direct). **Note:** one cost can be direct for one cost object and indirect for another.

3. Cost behaviour: variable, fixed, mixed

In simple words (layman view)

Variable = like a taxi fare: more km, more money, same rate per km. **Fixed** = like monthly rent: same total no matter how much you use, so the *per-unit* cost falls as you use more. Pizza example: a Rs 10 pizza cut into 4 slices costs 2.50 per slice; cut into 6, it costs 1.67 per slice; the total never changed.

	Total cost	Cost per unit
Variable cost	Changes **in proportion** with output (more output = more cost)	Constant
Fixed cost	Unchanged in total (within the relevant range)	Changes **inversely** with output (more output = lower cost per unit)

- Costs are fixed or variable only **with respect to a specific activity and time period**.
- **Mixed (semi-variable) costs** contain both parts, e.g. a utility bill = fixed monthly charge + rate per kWh. Others: advertising, phone, repairs. To budget/plan/control them we must **separate** the fixed and variable parts.

- **Types of fixed cost:** *Discretionary* (can be changed in the short term by managers: advertising, R&D) and *Committed* (long-term, cannot be cut quickly: depreciation on buildings, real estate taxes).
- **Relevant range** = the band of activity within which the fixed/variable assumptions hold; it is tied to capacity (a plane with 8 seats has a relevant range of 0 to 8 passengers).
- Understanding cost behaviour helps us **predict, estimate, budget, plan and control** costs.

Why "fixed cost per unit" needs care

Roommate example: rent is Rs 1,000 in total. 2 people = 500 each, 3 people = 333.33 each, 4 people = 250 each. Careful: seeing "500 per person" and multiplying by 4 to get 2,000 is wrong. Fixed costs are fixed **in total**, not per person.

4. Multiple classifications (BMW X6 style)

	Variable	Fixed
Direct	Direct materials (steel, tyres, steering wheels)	Salary of a supervisor dedicated to this one model
Indirect	Power used by machines (varies with output, but shared)	Plant rent, depreciation, plant administration

Exam tip / watch out

Unit-cost caveat. Unit costs (average costs) that include fixed costs are only valid for a given level of output. For most decisions, think in **total costs**, not unit costs.

5. Opportunity cost, sunk cost, relevance

- **Opportunity cost** = the net benefit of the path not taken (what you give up by choosing X).
- **Sunk cost** = already incurred, cannot be recovered, does not affect today's decision (may be a cost or expense).
- **Relevant information** has two features: (1) it occurs in the **future**, and (2) it **differs among alternatives**. Past (historical) costs are never relevant, although they may help in predicting.
- Key question: **"What difference will this action make?"** Revenues and costs that are the same in all alternatives are irrelevant.
- **Quantitative** factors can be measured in numbers; **qualitative** factors (satisfaction, morale) are hard to measure but just as important.
- Information for decisions should be **Relevant, Accurate and Timely**.
- **Decision model** = a formal method of choosing between courses of action (quantitative + qualitative analysis).

Term	Meaning	Example from slides
Differential cost	Amount by which cost differs between two alternatives	Commute cost 300 (city) vs 50 (hometown): differential = **250**
Differential revenue	Amount by which revenue differs	Salary 2,000 vs 1,500: differential = **500**
Marginal cost	Extra cost of producing **one more** unit	Mostly the variable cost
Average cost	Total cost / quantity produced	Includes fixed cost per unit

Worked mini-case (slides 41-42)

In the job example you gain 500 more salary but spend 250 more on commuting, so the move is better by **250** a month. Only the two *differences* matter. Everything else (what you ate, rent that stays the same) is irrelevant.

6. Answers to the class-participation questions

Question	Answer	Reason
QFT 4: Doctor says gasoline is fixed because cost per gallon is stable, insurance is variable because cost per mile falls as miles rise.	**Disagree on both**	Gasoline is **variable** (total cost rises with miles driven; stable price per gallon = constant per-unit variable cost). Insurance is **fixed** (total unchanged; cost per mile falls as miles rise, which is the signature of a fixed cost).
CP-3: Variable with number of ice-cream cones sold?	**C and D**	Ice cream and napkins vary with cones sold. Store lighting and manager wages are fixed.
CP-4: Total fixed cost of Elaine Co. shipping dept.	**(a) $850,000**	Salaries: 75% on guaranteed contracts = 0.75 x 800,000 = 600,000 fixed. Rent 250,000 fixed. Total = 850,000. Packaging and postage are variable.
CP-5: Rosland Graphics lost money at the same 0.27 price and same costs.	**(b)**	Costs were mostly fixed and this year's output was lower, so fixed cost per cover rose above 0.27.
CP-1: Is the train ticket cost relevant (drive vs train)?	**A: Yes**	Future cost that differs between alternatives.
CP-2: Is annual car licence cost relevant?	**B: No**	Paid either way; does not differ between alternatives.
CP-3 (b): Car could be sold now for $5,000. Sunk cost?	**B: No**	It is not yet spent: it is an **opportunity cost** (what you give up by keeping/using the car).

7. Types of firms and inventories

Sector	What they do	Inventory accounts	Examples
Manufacturing	Buy materials and convert them into finished goods	**Three**: Raw materials (direct materials), Work-in-process (WIP), Finished goods (FG); COGS on income statement	Microsoft, Boeing, P&G
Merchandising	Buy and resell goods without changing their form	**One**: Merchandise inventory	Walmart, grocery stores
Service	Provide intangible services	Usually none (no COGS); a "Cost of sales" line; small supply inventory possible	Audit firms, salons, FedEx

- **Direct materials**: resources in stock available for use. **WIP**: products started but not completed. **FG**: products completed and ready for sale.
- Costs flow: **Raw materials -> WIP -> Finished goods -> COGS** (balance sheet to income statement).

> **Flow of costs**
>
> Picture an orange-juice factory: oranges and cartons in the store room (RM inventory) -> machines mixing and bottling (WIP) -> cartons packed in the warehouse (FG) -> sold from the shop (COGS). Until a carton is sold, its cost sits on the **balance sheet**; the moment it is sold, the cost moves to the **income statement** as COGS.

Gross margin income statement (slide 50, with professor's notes): Sales - COGS (*manufacturing cost, MC*) = Gross margin; minus SG&A (*non-manufacturing cost, NMC, i.e. "period" costs / overheads*) = Operating income. Slide example: Sales 1,000 - COGS 410 = Gross margin 590; less SG&A 408 = Operating income 182. (COGS 410 = 250 variable + 160 fixed product costs; SG&A 408 = 270 variable + 138 fixed period costs.)

PART C: Day 3: Manufacturing Costs and the Flow of Costs

In simple words (layman view)

Making a product costs more than just the raw material. To know what a product *really* costs, add the material that goes into it, the wages of the people who make it, and all the other factory expenses (rent, power, supervisors...). Everything inside the factory gates is a **product cost**; everything outside (selling, office) is a **period cost**.

1. The three manufacturing (product / inventoriable) costs

Cost	Meaning	Examples
Direct Materials (DM)	Acquisition cost of all materials that become part of the product and can be traced to it	Steel in a car; oranges, water, cartons and lids in juice
Direct Labour (DL)	Wages, salaries and fringe benefits of workers who work directly on the product	Assembly-line worker's wages
Manufacturing Overhead (MOH)	All other factory costs: indirect material + indirect labour + other indirect costs	Lubricants, cleaning supplies, factory supplies (hair nets, gloves); maintenance staff, janitors, security guards, supervisors; factory depreciation, rent, property tax, insurance, utilities, repairs, overtime premium, unavoidable idle time

- **Indirect material**: needed for production but not an integral part of the product (or too small in value to trace).
- **Indirect labour**: people who support but do not work directly on the product.
- **Other indirect costs**: everything else in the factory that is neither material nor labour.

Prime cost = DM + DL (the direct costs). **Conversion cost = DL + MOH** (the cost of converting materials into finished goods). Memory trick: DL appears in both groups. Clay-mug example: clay (DM) + potter (DL) + kiln (MOH).

In simple words (layman view)

Prime = "the main ingredients you can point at". Conversion = "what it costs to turn the ingredients into something". The professor's margin note: DM and DL are **Direct**; MOH is **Indirect**. Within MOH, costs may be fixed (F) or variable (V).

2. Non-manufacturing costs (period costs)

- **Selling costs**: costs necessary to secure the order and deliver the product (advertising, sales commission, freight outward).
- **Administrative costs**: all executive, organisational and clerical costs (office salaries, office depreciation).
- Incurred by **all** types of businesses; "the cost of being in business". What is selling vs administrative can be "a matter of choice".

	Product costs	Period costs
Include	DM + DL + MOH	All selling and administrative costs
Where first recorded	Inventory (**balance sheet**)	Expense (**income statement**) straight away
When expensed	When the product is sold, as **COGS**	In the period incurred
Also called	Manufacturing / inventoriable costs	Non-manufacturing costs / NMC / operating expenses

Exam tip / watch out

Closing inventory still on hand carries its product cost on the **balance sheet**. Only the portion sold becomes COGS on the income statement. A single building used for manufacturing, selling and admin must **not** be expensed in one go (see the Madi Corporation answer, Question for Thought 5, below).

3. Professor's "factory funnel" method: how costs flow

The professor draws a funnel with **RM/DM, DL and MOH** flowing in. Whatever goes into the funnel in the period is the **Total Manufacturing Cost (TMC) = Cost of Production (COP) gross**. Then "switch on" three inventory stages, always asking: **Opening + Added = Available (AFU / AFS); Available - Closing = Used/Sold**.

Stage	Opening + Added	= Available	- Closing	= Used / Sent on
1. Direct materials (DM)	Opening + purchases + freight - returns	DM AFU (available for use)	Closing DM	**DM used** (goes into funnel)
2. Funnel	DM used + DL + MOH	= **TMC / COP gross**		
3. Work in process (WIP)	Opening WIP + TMC	WIP AFU	Closing WIP	**WIP used = COGM** (cost of goods manufactured / "FG today" / net COP)
4. Finished goods (FG)	Opening FG + COGM	FG AFS (available for sale; "COGM AFS")	Closing FG (product cost, **B/S**)	**COGS** (period expense, **P&L**)

Professor's vocabulary

Vocabulary in one line: AFU = Available For Use (materials/WIP stages); **AFS** = Available For Sale (FG stage). "COP gross" = TMC = costs added this period before the WIP adjustment. "COGM / COP net" = after WIP adjustment. Closing stocks at each stage are **product costs** (balance sheet); only COGS and the non-manufacturing costs hit the **P&L**.

Worked example from class: Day 1

Direct materials	Rs	Funnel (Day 1)	Rs
Opening	0	DM used	200
+ Purchase	400	DL	100
+ Freight	50	MOH	50
- Return	(25)	**COP gross = TMC**	**350**
DM AFU	**425**		
- Closing	(225)		
DM used	**200**		

Work in process	Rs	Finished goods	Rs
Opening WIP	0	Opening FG	0
+ Added (TMC)	350	+ Added (COGM)	275
WIP AFU	**350**	**FG AFS**	**275**
- Completed (WIP used = COGM)	(275)	- Closing FG (B/S, product cost)	(50)
Closing WIP	**75**	**COGS (P&L, period)**	**225**

Worked example from class: Day 2 (opening balances carried forward)

Direct materials	Rs	Funnel (Day 2)	Rs
Opening (Day 1 closing)	225	DM used	300
+ Purchase	400	DL	200
+ Freight	100	MOH	100
- Return	(25)	**COP gross = TMC**	**600**
DM AFU	**700**		

Direct materials	Rs	Funnel (Day 2)	Rs
- Closing	(400)		
DM used	**300**		

Work in process	Rs	Finished goods	Rs
Opening WIP (Day 1 closing)	75	Opening FG (Day 1 closing)	50
+ Added (TMC)	600	+ Added (COGM)	400
WIP AFU	**675**	**FG AFS**	**450**
- Completed (COGM)	(400)	- Closing FG (B/S)	(75)
Closing WIP	**275**	**COGS (P&L)**	**375**

In simple words (layman view)

Notice how each day's **closing** becomes the next day's **opening**. This is the heart of the method: nothing is lost, every rupee is either used, sold, or still sitting in some inventory.

4. Income statement layout (professor's order)

Line	Notes
Sales	
(-) COGS	Manufacturing cost (MC) of what was sold
= Gross margin (GM)	
(-) Sales (selling) expenses	Non-manufacturing cost (NMC) / period cost
(-) Admin expenses	Also NMC. Professor's note: in the Cost Sheet, admin overhead may instead be added to cost of production (see Set 1, Problem 1 alternative)
= Operating profit (OP)	

Two formats appear in the course: the **Gross margin statement** (by function: product vs period) above, and the **Contribution margin statement** (by behaviour: variable vs fixed): Sales - all variable costs (product + period) = Contribution margin; less all fixed costs = Operating income. Slide example: variable 250 (product) + 270 (period); fixed 160 (product) + 138 (period).

Exam tip / watch out

Whether admin overhead is added to the cost of production (traditional cost sheet) or kept as a period cost (GAAP/management accounting income statement) changes COGS and closing-inventory values but **not** the cost sheet's total cost. Always say which method you are using.

5. Now You Do It: Solomon Grundy Corp.

Facts: no beginning inventories. Tue: bought and paid cash for 10,000 raw materials (all direct). Wed: removed 3,600 DM and started production. Thu: DL 2,500. Fri: MOH 800. Sat: finished 80% of production. Sun: sold 70% of what was finished.

Day	Transaction	Effect
Tue	Purchase of RM	RM inventory = 10,000
Wed	DM to production	RM = 6,400 ; WIP = 3,600
Thu	Direct labour	WIP = 3,600 + 2,500 = 6,100
Fri	MOH	WIP = 6,100 + 800 = 6,900
Sat	Finish 80%	COGM = 0.8 x 6,900 = **5,520** to FG; WIP left = **1,380**
Sun	Sell 70% of FG	COGS = 0.7 x 5,520 = **3,864**; FG left = **1,656**

- **(b) Saturday balances:** RM = 6,400; WIP = 1,380; FG = 5,520.
- **(c) Sunday COGS = 3,864.**

6. Answer to Question for Thought 5 (Madi Corporation)

Answer

One building is used for manufacturing, selling and administration. Its depreciation should **not** be dumped entirely into one expense. Split it (e.g. by floor area used) among the three functions: the **factory share** is manufacturing overhead (a **product cost**, held in inventory until sold), while the **selling and administrative shares** are **period costs** expensed immediately. This is exactly what the professor did in Set 1, Problem 1: building depreciation split 8:1:1 between factory, office and selling.

PART D: Day 4: Overhead Accounting, Analysis and Control

In simple words (layman view)

Overheads are the "shared" costs (rent, power, supervisors, canteen). You cannot point at a single product and say "this much of the rent is yours", so we **share them out in a fair way**: first to departments, then from helper (service) departments to the departments that actually make the product, and finally to each product.

1. Key definitions

Term	Meaning
Overheads	Any expenditure over and above prime cost = indirect material + indirect labour + indirect expenses
Cost allocation	Charging a **whole, identifiable** cost item to one cost centre or cost unit (e.g. a supervisor who works only in Dept A is charged fully to A)
Cost apportionment	Dividing a cost among **two or more** cost centres in proportion to estimated benefit received (e.g. factory rent shared by area)
Absorption	Charging overhead to the products (cost units) using a pre-determined rate
Operating / production department	Directly adds value to the product or service
Support / service department	Provides services to other departments (stores, maintenance, canteen, accounts, inspection, purchasing)

Classification of overheads: by **function** (Production OH, Admin OH, Selling & Distribution OH); by **element** (indirect material, indirect labour, indirect expenses); by **behaviour** (variable, fixed, semi-variable).

2. Production overhead: five steps

1. **Classification** of overheads.
2. **Collection** under standing order numbers (documents: stores requisitions, time cards/wages analysis sheets, invoices/vouchers, cash book for petty expenses, subsidiary records for non-cash items like depreciation and notional rent).
3. **Distribution** to production and service departments: **Primary distribution** (departmentalisation).
4. **Redistribution** of service department costs to the production departments: **Secondary distribution**.
5. **Absorption** of overhead by production units using predetermined overhead rates.

Primary distribution: common bases (cost drivers)

Item of expenditure	Basis of distribution
Rent, depreciation, insurance (building)	Area occupied (floor area)
Insurance and depreciation of plant & machinery	Capital value (value of plant)
Power	Horse power x machine hours (or direct if metered)
Heating & lighting	Floor area / light points
Electricity	Metered consumption directly; else light points
Supervision, employer's insurance, canteen, time-keeping	Number of employees / wages
Fire insurance on stock	Value of stock

Selecting an allocation base: three criteria: **simplicity**, **availability** of space/equipment data, and **benefits received** by the receiving department. Typical bases: Personnel = number of employees; Receiving = units handled; Security/Custodial = square footage; Power = kWh; Cafeteria = employees; Accounting = staff hours. The rate = accumulated cost (numerator) / total base (denominator).

3. Secondary distribution: bases for service departments

- Inspection: inspection hours, number of inspectors, pieces
- Purchase: number of orders, quantity or value
- Repairs: hours worked; asset value x hours worked
- Time-keeping: number of employees, labour hours, machine hours
- Accounts: number of workers
- Stores: value or number of stores requisitions

> **In simple words (layman view)**
>
> Service department costs are not charged to products directly. They are first **pushed into the production departments**, become part of each department's overhead, and then reach products through the **predetermined overhead rate**. (Slide flow: first-stage allocation = service to production departments; second-stage = production department OH + allocated service OH applied to products.)

4. Single-rate vs dual-rate allocation

- **Single-rate method**: does not separate fixed and variable costs; one rate per unit of a single allocation base for each cost pool (each support department).
- **Dual-rate method**: splits each support department's cost into a **variable pool** and a **fixed pool** and allocates each with a different base. More accurate; can be combined with the reciprocal method.

5. Three methods for allocating service department costs

Method	How it treats services between support departments	Accuracy / use
Direct	Ignores them completely. Each service dept's cost goes straight to production departments.	Simplest; widely used
Step-down (step)	Recognises them **partially**: one-way. Start with the dept that serves the most other service depts; once allocated, never charge back to it.	More precise than direct; sequence affects answers
Reciprocal	Recognises them **fully**: two-way. Needs simultaneous equations (best done in Excel) or repeated distribution.	Most precise; rarely used, results usually close to step method

5a. Direct (redistribution) method: steps

1. Take each service department's total cost.
2. Ignore any service one service dept gives to another.
3. Find the ratio of the base only among the **production** departments (simplify the ratio).
4. Multiply each share by the cost.
5. Add to the production departments' primary-distribution totals; check the grand total is unchanged.

5b. Step-down method: steps

1. Rank service departments: start with the one that serves the largest number of other service departments (or has the highest cost).
2. Allocate the first department's cost to **all other departments** (service + production) using its base.
3. Move to the next: its new total = own cost + cost received. Allocate to the **remaining** service departments and production departments only (never back).
4. Continue until all service costs are in production departments.

5c. Reciprocal method: two ways

- **Repeated distribution (cycles)**: keep re-distributing service totals back and forth by agreed percentages until the amounts become too small to matter.
- **Simultaneous equations**: write each service department's true cost as an equation including the share it gets back, solve, then distribute the true costs to production departments by the given percentages.

6. Administration overhead and selling & distribution overhead

- **Administration OH** relates to the business as a whole (board, legal, accounting, audit, personnel, finance). Three treatments: (1) **transfer to P&L** if fixed by nature with no link to production/sales; (2) **apportion** to production and sales functions on an equitable basis; (3) treat as a **separate addition** to cost of production/sales, usually based on works cost (best), sales value, gross profit, quantity or conversion cost.
- **Selling & distribution OH** can be analysed by nature (freight, duty, packing, promotion, discounts), by function (transportation, direct selling, advertising, credit, storage), by area, product, salesman, type of customer, channel. **Control** through comparison with past results, budgetary control and standard costing.

7. Allocating common costs

In simple words (layman view)

A **common cost** is shared by two or more users because sharing is cheaper than each paying separately (like two siblings sharing one phone plan). The question is how to split the bill fairly.

Method	How it works
Stand-alone	Find what each user would pay **if it paid alone**; allocate the actual common cost in **proportion** to those stand-alone costs. Costs are shared proportionally, not equally.
Incremental	Rank users. The **primary user** is charged its stand-alone cost; the next user is charged the **additional** cost caused by adding it; and so on. Ranking matters (primary user bears the most).

Slide example (phone plan): brother alone = 75, you alone = 85, shared plan = 115.

	Brother	You	Total
Stand-alone cost	75	85	160
Share % (75/160; 85/160)	47%	53%	100%
Stand-alone allocation of 115	54.05 (47% x 115)	60.95 (53% x 115)	115
Incremental (brother primary)	75	40 (115 - 75)	115

Exam tip / watch out

The slide rounds the shares to whole percentages (47% / 53%). Using exact fractions gives 46.875% x 115 = 53.91 and 53.125% x 115 = 61.09. Both approaches are shown in textbooks; use the rounding your professor specifies.

PART E: Problem Sets: Solutions in the Professor's Method, with Excel Steps

In simple words (layman view)

Every cost problem is the same story told in a fixed order. Learn the order once and all problems become easy: **(1) Prime cost -> (2) add factory overhead -> (3) adjust for opening/closing WIP -> (4) adjust for opening/closing FG -> (5) add non-manufacturing costs -> (6) compare with sales.**

The master skeleton (use for every problem)

Step	Name used by professor	Formula
1	DM AFU (available for use)	Opening DM + Purchases + Freight - Returns
2	DM used	DM AFU - Closing DM
3	Prime cost (PC)	DM used + DL (+ direct expenses)
4	Factory OH (MOH)	Indirect material + indirect labour + other factory costs + factory share of depreciation
5	Works cost gross = COP gross = TMC	PC + MOH
6	WIP adjustment -> WIP AFU	TMC + Opening WIP
7	Works cost net = COGM (WIP used)	WIP AFU - Closing WIP
8	FG AFS (COG available for sale)	COGM + Opening FG
9	COGS (FG used)	FG AFS - Closing FG
10	Non-manufacturing (period) costs	Administration OH + Selling & distribution OH
11	Total cost / cost of sales	COGS + non-manufacturing costs
12	Profit	Net sales - total cost (Gross margin = Sales - COGS; Operating profit = GM - NMC)

Exam tip / watch out

Back-solving (Set 2) just runs the same skeleton backwards: when a closing balance is unknown, find it as **Available - Used**; when "used" is unknown, find it as **Available - Closing**.

Excel basics used in all problems

1. **Type labels in column A, numbers in column B**. Never type a calculated number: always type a formula starting with **=**.
2. **Add / subtract:** `=B2+B3-B4`; **total a range:** `=SUM(B2:B6)`.
3. **Percentages:** `=B10*0.4` or format a cell as % and use `=B10*C10`.
4. **Lock a cell with $** (absolute reference) when copying a formula: `$B$2` always points at B2; `C$5` locks only the row; `$B7` locks only the column. Press **F4** to cycle through these.
5. **Copy a formula:** select the cell, drag the small square at its bottom-right corner across/down.
6. **Share by ratio:** `=C5/SUM($C$5:$F$5)*$B7` (this cell's base / total base x cost). **Check:** total of the shares must equal the original cost.
7. **Sum of products:** `=SUMPRODUCT(range1, range2)`.
8. **Rounding:** `=ROUND(B5,0)`. **Checks:** add a check cell, e.g. `=D20-SUM(D5:D8)`, which should show 0.
9. **Solving unknowns:** Data -> What-If Analysis -> **Goal Seek**, or rearrange the equation algebraically on paper first (as the professor does for CC = 18/0.6).
10. **Format:** Ctrl+1 -> Number -> use thousands separator; **Ctrl+B** for bold totals; **Home -> Borders** for total lines.

SET 1: Cost sheets (Problem Set 1)

Set 1, Problem 1: Cost sheet and profit/loss of ABC Ltd (year ended 31.03.2013)

In simple words (layman view)

We must find out how much it cost ABC to make and sell its goods, then compare with sales. Items that are not part of operating the business (interest on borrowings, bad debts, dividend income) are **financial items** and are left out of the cost accounts.

Professor's method, step by step

Step 1: DM used. Gross purchases 320,000 + freight 16,000 - purchase returns 4,800 (a credit, so it reduces) = 331,200. Add opening stock 140,000 = **DM AFU 471,200**. Less closing stock 180,000 = **DM used 291,200**.

Step 2: DL = actual 160,000 + accrued 8,000 = **168,000**. **Direct expense** = 50,000.

Step 3: Prime cost = 291,200 + 168,000 + 50,000 = **509,200**.

Step 4: Factory OH (see table). Split common expenses as told: heat/light/power 8:1:1 (factory : office : selling); rates & taxes 2/3 factory, 1/3 office; building depreciation 8:1:1.

Factory overhead item	Working	Rs
Indirect labour	18,000 + 1,200 accrued	19,200
Factory supervision		10,000
Factory repairs and upkeep		14,000
Heat, light and power	8/10 x 65,000	52,000
Rates and taxes	2/3 x 6,300	4,200
Miscellaneous factory expenses		18,700
Depreciation: plant & machinery	10% x 460,500	46,050
Depreciation: buildings (factory share)	4% x 200,000 x 8/10	6,400
Total factory overhead		**170,550**

Stage	Working	Rs
Prime cost	291,200 + 168,000 + 50,000	509,200
+ Factory overhead		170,550
Works cost gross = COP gross = TMC		**679,750**
+ Opening WIP		200,000
WIP AFU		879,750
- Closing WIP		(192,000)
Works cost net = COGM	**WIP used / FG made today**	**687,750**
+ Opening FG		80,000
FG AFS		767,750
- Closing FG		(115,000)
COGS	**FG used**	**652,750**

Non-manufacturing (period) costs	Working	Rs
Administration OH: heat, light, power	1/10 x 65,000	6,500
Rates and taxes	1/3 x 6,300	2,100
Office salaries and expenses		8,600
Depreciation: building (office)	1/10 x 8,000	800
Depreciation: office appliances	5% x 17,500	875
Total administration OH		18,875
Selling & distribution OH: heat, light, power	1/10 x 65,000	6,500
Sales commission		33,600

Non-manufacturing (period) costs	Working	Rs
Sales travelling		11,000
Sales promotion		22,500
Distribution salaries & expenses		18,000
Depreciation: building (selling)	1/10 x 8,000	800
Total selling & distribution OH		92,400
Total non-manufacturing cost	**18,875 + 92,400**	**111,275**

Profit and loss as per cost accounts	Rs
Sales	768,000
Less: sales returns and rebates	(14,000)
Net sales	**754,000**
Less: COGS	(652,750)
Gross margin	**101,250**
Less: non-manufacturing cost	(111,275)
Operating result: LOSS	**(10,025)**

Two ways, same answer

Alternative presentation (the solution file's cost sheet): add administration OH (18,875) to works cost net (687,750) = **cost of production 706,625** (the solution file prints 706,620, a typo). Then +80,000 - 115,000 = 671,625 (cost of production available for sale) + selling & distribution OH 92,400 = **cost of sales 764,025**; net sales 754,000 - 764,025 = **loss 10,025**. Both methods give the same loss; only where admin cost sits differs.

Exam tip / watch out

Excluded from the cost accounts: interest on borrowed funds (2,000 + 2,000 accrued), bad debts (5,000) and dividend income (20,000). They are financial items. Accrued DL and IL are **added** to the actual amounts. Purchase returns are a **credit** (they reduce purchases).

How to do it in Excel

Layout: open a sheet, put labels in column A, values in column B (like the professor's "P1s1" sheet). Suggested cell map:

Cell	Content	Formula / value
B2:B6	Purchases, freight, returns, opening RM, closing RM	320000, 16000, 4800, 140000, 180000
B7	DM AFU	=B2+B3-B4+B5
B8	DM used	=B7-B6
B9	Direct labour (actual+accrued)	=160000+8000
B10	Direct expense	50000
B11	PRIME COST	=B8+B9+B10
B13	Factory OH	=19200+10000+14000+0.8*65000+(2/3)*6300+18700+0.1*460500+0.04*200000*0.8
B14	WORKS COST GROSS (TMC)	=B11+B13
B15	WIP AFU	=B14+200000
B16	COGM (works cost net)	=B15-192000
B17	FG AFS	=B16+80000
B18	COGS	=B17-115000
B20	Admin OH	=0.1*65000+(1/3)*6300+8600+0.04*200000*0.1+0.05*17500
B21	Selling & dist. OH	=0.1*65000+33600+11000+22500+18000+0.04*200000*0.1
B22	Total non-mfg cost	=B20+B21
B24	Net sales	=768000-14000
B25	Gross margin	=B24-B18
B26	Operating profit / (loss)	=B25-B22

Steps: (1) type each input in its own cell instead of embedding numbers (better practice: put 65000 in a cell and use `=0.8*$B$30`). (2) Build formulas top to bottom exactly as the skeleton. (3) Press Enter, check B11 = 509,200, B14 = 679,750, B16 = 687,750, B18 = 652,750, B26 = -10,025. (4) Format negative numbers in brackets: Ctrl+1 -> Number -> Custom -> `#,##0;(#,##0)`.

Set 1, Problem 2: Cost statement, estimated cost and price (Steelsways Engineering)

In simple words (layman view)

First, use last year's accounts to find how big overheads are *relative* to something simple (wages, works cost). Then use those percentages to estimate the cost of a new machine and add the profit to get the quote price.

Professor's method

(a) Works cost = materials 180,000 + wages 160,000 + works overhead 40,000 = **380,000**. Total cost = works cost + establishment & general 19,000 = **399,000**.

Works overhead % of wages = 40,000 / 160,000 = **25%**. Establishment & general % of works cost = 19,000 / 380,000 = **5%**.

(b) Estimate: materials 8,000 + wages 6,000 = prime cost 14,000. Works overhead 25% of wages = 1,500. Works cost = 15,500. Establishment 5% of works cost = 775. **Total cost = 16,275.**

Profit 25% on cost = 4,068.75, so price = **20,343.75 (about 20,344)**. Check with **20% on selling price**: price = cost / 0.80 = 16,275 / 0.8 = 20,343.75. Same, because 25% on cost = 20% on price.

Exam tip / watch out

"Profit x% on cost" -> price = cost x (1 + x). "Profit y% on selling price" -> price = cost / (1 - y). 25% on cost is identical to 20% on price (1.25 = 1/0.8).

How to do it in Excel

Cell	Item	Formula
B2:B4	Materials, wages, works OH	180000, 160000, 40000
B5	Works cost	=B2+B3+B4
B6	E&G expenses	19000
B7	Total cost	=B5+B6
B8	Works OH % of wages	=B4/B3 (format as %)
B9	E&G % of works cost	=B6/B5 (format as %)
B11:B12	New machine: materials, wages	8000, 6000
B13	Works OH	=B12*B8
B14	Works cost	=B11+B12+B13
B15	E&G	=B14*B9
B16	Total cost	=B14+B15
B17	Profit 25% on cost	=B16*0.25
B18	QUOTE PRICE	=B16+B17 (check: =B16/(1-0.2))

Set 1, Problem 3: Cost sheet with inventory adjustments (year ended 31.03.2013)

Professor's method

Materials consumed = purchases 110,000 + opening 3,000 - closing 4,000 = 109,000. **Prime cost** = 109,000 + DL 65,000 + chargeable expenses 10,000 = **184,000**.

Factory OH = 50% of DL = 32,500, so TMC (works cost gross) = **216,500**.

WIP adjustment: + opening 4,000 - closing 6,000 = -2,000, so **works cost net = 214,500**.

Admin OH = 10% of works cost = 21,450. Cost of goods manufactured (traditional) = **235,950**.

FG adjustment: + opening 8,000 - closing 7,000 = +1,000, so **cost of goods sold = 236,950**.

Selling OH = 5% of sales 275,000 = 13,750. **Cost of sales = 250,700**. **Profit = 275,000 - 250,700 = 24,300.**

In simple words (layman view)

Alternative (admin treated as a period cost): COGM = works cost net 214,500; +8,000 - 7,000 -> COGS = **215,500**; add non-manufacturing: selling 13,750 + admin 21,450 -> cost of sales **250,700**; profit **24,300**. Same profit.

How to do it in Excel

Cell	Item	Formula
B2:B4	Purchases, opening RM, closing RM	110000, 3000, 4000
B5	Materials consumed	=B2+B3-B4
B6:B7	DL, chargeable expenses	65000, 10000
B8	Prime cost	=B5+B6+B7
B9	Factory OH (50% DL)	=0.5*B6
B10	Works cost gross	=B8+B9
B11	Works cost net	=B10+4000-6000
B12	Admin OH (10% works cost)	=0.1*B11
B13	COGM	=B11+B12
B14	COGS	=B13+8000-7000
B15	Selling OH (5% of sales)	=0.05*275000
B16	Cost of sales	=B14+B15
B17	Profit	=275000-B16

Set 1, Problem 4: Cost of sales and profit determination (four weeks ended 30 June 2013)

Professor's method

Prime cost = raw materials 4,000 + wages 6,000 = 10,000 (0.50 per unit on 20,000 units).

Works overhead = machine hours 1,000 x 0.50 = 500 (0.025 per unit). **Works cost** = 10,500 (0.525 per unit).

Office overhead = 20% of works cost = 2,100 (0.105 per unit). **Cost of production** (20,000 units) = 12,600, i.e. **0.63 per unit**.

Closing stock = 20,000 produced - 18,000 sold = 2,000 units x 0.63 = 1,260. **COGS** (18,000 units) = 11,340.

Selling overhead = 18,000 x 0.06 = 1,080. **Cost of sales** = 12,420 (0.69 per unit).

Sales = 18,000 x Re 1 = 18,000. **Profit = 5,580** (0.31 per unit).

Why two answers?

Alternative (office overhead as period cost): closing stock = 2,000 x 0.525 = 1,050; COGS = 9,450; non-manufacturing = selling 1,080 + office 2,100 = 3,180; cost of sales = **12,630**; profit = **5,370**. The profit differs (5,580 vs 5,370) because in the first method part of the office overhead (2,000 x 0.105 = 210) is carried in closing stock instead of being expensed.

How to do it in Excel

Cell	Item	Formula
B2:B3	Units produced, units sold	20000, 18000
B4:B5	Raw materials, wages	4000, 6000
B6	Prime cost	=B4+B5
B7	Works OH	=1000*0.5
B8	Works cost	=B6+B7
B9	Office OH (20%)	=0.2*B8
B10	Cost of production	=B8+B9
B11	Cost per unit	=B10/B2
B12	Closing stock units / value	=B2-B3 / =B12*B11
B14	COGS	=B10-closing stock value
B15	Selling OH	=B3*0.06
B16	Cost of sales	=B14+B15
B17	Sales / Profit	=B3*1 / =sales-B16

SET 2: Back-solving inventories and absorption costing (Problem Set 2)

In simple words (layman view)

In these problems some inventory numbers are hidden. Use the "**Opening + Added = Available; Available - Closing = Used**" box at every stage and work **backwards from what you know** (the professor calls these "stage 1, stage 2, stage 3").

Set 2, Q1: Fire at the plant (Mr. Unsatisfied): find FG, WIP and DM stock on 28 Feb

Given (Rs lakh): DM purchased 16; WIP 1 Jan 3.4; DM 1 Jan 1.6; FG 1 Jan 3.0; Revenue 50; DL 18; Prime cost (PC) 29.40; indirect factory costs = 40% of conversion cost; COG available for sale 45; gross margin 20% of revenue.

Professor's method

Stage 1: direct materials. DM AFU = 1.6 + 16 = **17.6**. PC = DM used + DL = 29.4, so DM used = 29.4 - 18 = **11.4**. Closing DM = 17.6 - 11.4 = **6.2**.

Find MOH. Conversion cost CC = DL + MOH and MOH = 0.4 CC, so DL = 0.6 CC. Therefore 18 = 0.6 CC, **CC = 18 / 0.6 = 30**, **MOH = 12** (check: 18 + 12 = 30).

Stage 2: funnel. COP gross (TMC) = PC + MOH = 29.4 + 12 = **41.4**. WIP AFU = 3.4 + 41.4 = **44.8**.

Stage 3: finished goods (work backwards from the income statement). Gross margin 20% of 50 = 10, so COGS = 50 - 10 = **40**. FG AFS = 45 (given). Closing FG = 45 - 40 = **5**.

Back to WIP. FG AFS = opening FG 3 + COGM, so COGM = 45 - 3 = **42** (this is the WIP used). Closing WIP = WIP AFU - WIP used = 44.8 - 42 = **2.8**.

Stock on 28 Feb	Working	Rs lakh
DM stock	17.6 - 11.4	**6.2**
WIP stock	44.8 - 42	**2.8**
FG stock	45 - 40	**5.0**

How to do it in Excel

Cell	Item	Formula
B2:B7	Inputs: DM purchased, opening DM, opening WIP, opening FG, DL, PC	16, 1.6, 3.4, 3.0, 18, 29.4
B8:B10	Revenue, GM %, COG AFS	50, 0.2, 45
B12	DM AFU	=B2+B3
B13	DM used	=B7-B6
B14	Closing DM	=B12-B13
B15	Conversion cost CC	=B6/0.6
B16	MOH	=B15-B6
B17	COP gross (TMC)	=B7+B16
B18	WIP AFU	=B17+B4
B19	COGS	=B8*(1-B9)
B20	Closing FG	=B10-B19
B21	COGM (WIP used)	=B10-B5
B22	Closing WIP	=B18-B21

Change any input (e.g. revenue) and every answer updates: that is the advantage of formulas.

Set 2, Q2: ABC metal shelving: DM stock, FG units, selling price, operating income

Given: costs in Rs lakh: DM 1.4 V; DL 0.3 V; plant energy 0.05 V; indirect labour 0.10 V + 0.16 F; indirect other 0.08 V + 0.24 F; marketing/distribution/customer service 1.2285 V + 0.4 F; admin 0.5 F. Variable manufacturing costs vary with units **produced**; variable marketing costs vary with units **sold**. Production = 100,000 units (1 lakh); 2 kg DM per unit; DM stock 0 -> 2,000 kg; no WIP; revenue Rs 436,800; closing FG is valued at the average manufacturing cost per unit and equals Rs 20,970.

Professor's method

Step 1: DM. DM used = 100,000 units x 2 kg = 200,000 kg. Purchased = 200,000 + 2,000 closing = 202,000 kg. The Rs 1.4 lakh is the cost of DM **used** (consumed), i.e. 140,000 / 200,000 = Rs 0.70 per kg. **Closing DM = 2,000 kg x 0.70 = Rs 1,400.** (This reading is confirmed in step 3: it gives a whole number of FG units.)

Step 2: total manufacturing cost (all in Rs): DM 140,000 + DL 30,000 + plant energy 5,000 + indirect labour (10,000 + 16,000) + indirect other (8,000 + 24,000) = **233,000**. Per unit = 233,000 / 100,000 = **Rs 2.33**.

Step 3: FG units. 20,970 / 2.33 = **9,000 units**. Units sold = 100,000 - 9,000 = **91,000**.

Step 4: selling price = 436,800 / 91,000 = **Rs 4.80 per unit**.

Step 5: COGS = 233,000 - 20,970 = **212,030**. Check: 91,000 x 2.33 = 212,030.

Step 6: marketing costs. Variable = 1.2285 lakh = 122,850 (= 91,000 x 1.35). Fixed = 40,000. Admin (fixed) = 50,000.

Income statement (Rs)	Rs
Revenue (91,000 x 4.80)	436,800
Less: COGS (91,000 x 2.33)	(212,030)
Gross margin	**224,770**
Less: marketing, distribution & customer service: variable	(122,850)
Less: marketing, distribution & customer service: fixed	(40,000)
Less: administration (fixed)	(50,000)
Operating income	**11,920**

Answers	
Period-ending DM inventory cost	Rs 1,400 (2,000 kg)

Answers	
Period-ending FG inventory	9,000 units
Selling price per unit	Rs 4.80
Operating income	Rs 11,920

Exam tip / watch out

Fixed manufacturing costs (here 16,000 + 24,000 = 40,000) are **absorbed into inventory** in absorption costing: that is why FG carries 2.33 per unit and not only variable costs. Marketing and admin costs are **period costs**; they never enter inventory.

How to do it in Excel

Cell	Item	Formula
B2:B3	Units produced, kg per unit	100000, 2
B4	Closing DM kg	2000
B5	DM kg used	=B2*B3
B6	Cost per kg	=140000/B5
B7	**Closing DM cost**	=B4*B6
B9	Total mfg cost	=140000+30000+5000+10000+16000+8000+24000
B10	Mfg cost per unit	=B9/B2
B11	**FG units**	=20970/B10
B12	Units sold	=B2-B11
B13	**Selling price**	=436800/B12
B14	COGS	=B9-20970
B15	Variable marketing	=122850
B16	**Operating income**	=436800-B14-B15-40000-50000

Set 2, Q3: Indirect other manufacturing costs are not inventoriable; 9,000 units of FG on 31 Dec

In simple words (layman view)

If a cost is declared "not inventoriable", it is treated as a **period cost**: expensed in the year, never put into closing stock. So closing FG carries **less** cost per unit and profit falls.

Professor's method

Indirect other (8,000 variable + 24,000 fixed = **32,000**) is expensed immediately.

Inventoriable manufacturing cost = 233,000 - 32,000 = **201,000**; per unit = 201,000 / 100,000 = **Rs 2.01**.

FG ending cost = 9,000 x 2.01 = **Rs 18,090**. **COGS** = 201,000 - 18,090 = **182,910**.

Income statement (Rs)	Rs
Revenue	436,800
Less: COGS	(182,910)
Gross margin	**253,890**
Less: indirect other manufacturing (period cost)	(32,000)
Less: marketing variable + fixed (122,850 + 40,000)	(162,850)
Less: administration	(50,000)
Operating income	**9,040**

Check: operating income in Q2 (11,920) minus Q3 (9,040) = 2,880 = 9,000 units x (2.33 - 2.01) = the cost that is no longer carried in closing FG.

How to do it in Excel

Cell	Item	Formula
B2	Inventoriable mfg cost	=233000-32000
B3	Cost per unit	=B2/100000
B4	**FG ending cost**	=9000*B3
B5	COGS	=B2-B4
B6	**Operating income**	=436800-B5-32000-162850-50000

Set 2, Q4: ABC, September 2014 (Rs million)

Given: opening WIP 2,000; opening DM 900; DM purchased 3,600; DM used 3,750; variable mfg OH 2,500; total mfg OH 4,800; total manufacturing costs (TMC) 16,000; COGM 16,500; COGS 17,000; closing FG 1,250.

Professor's method

DM: AFU = 900 + 3,600 = 4,500. Closing DM = 4,500 - 3,750 = **750**.

DL: TMC = DM used + DL + MOH, so DL = 16,000 - 3,750 - 4,800 = **7,450**.

Fixed MOH = total 4,800 - variable 2,500 = **2,300**.

WIP: AFU = opening 2,000 + TMC 16,000 = 18,000; COGM (WIP used) = 16,500; closing WIP = 18,000 - 16,500 = **1,500**.

FG: goods available for sale = COGS + closing FG = 17,000 + 1,250 = **18,250**. Opening FG = 18,250 - COGM 16,500 = **1,750**.

Answer	Rs million
DM inventory 30 Sept	750
WIP inventory 30 Sept	1,500
FG inventory 30 Sept	1,250 (given)
Fixed manufacturing OH	2,300
Direct labour	7,450
Goods available for sale	18,250 (opening FG 1,750 + COGM 16,500)

How to do it in Excel

Cell	Item	Formula
B2:B5	Open DM, purchases, DM used, TMC	900, 3600, 3750, 16000
B6	Closing DM	=B2+B3-B4
B7	DL	=B5-B4-4800
B8	Fixed MOH	=4800-2500
B9	Closing WIP	=2000+B5-16500
B10	Goods available for sale	=17000+1250

SET 3: Overhead distribution (Problem Set 3)

In simple words (layman view)

Think of a hostel bill: the electricity bill is shared by room (area), the mess bill by number of residents, the lift charge by floors. Choose the **fairest basis** for each cost, turn it into a ratio, simplify it, and split the cost. Always check that the shares add back to the original total.

Set 3, Problem 1: Primary distribution (departments A, B, C production; D service)

Costs (Rs): Rent 10,000; repairs to plant 6,000; depreciation of plant 4,500; supervision 15,000; fire insurance 5,000; power 9,000; light 1,000; employer's insurance liability 1,500 (total 52,000).

Professor's method (ratio method)

Step 1: pick the basis that best matches what causes each cost (area for rent, value of plant for repairs/depreciation, wages for supervision, value of stock for fire insurance, HP of machines for power, light points for light, number of employees for employer's insurance).

Step 2: write the ratio of the base across A : B : C : D, strike common zeros, and divide by the common factor. Example: area 1500 : 1100 : 900 : 500 -> 15 : 11 : 9 : 5 (total 40). Plant value 240,000 : 180,000 : 120,000 : 60,000 -> 24 : 18 : 12 : 6 -> 4 : 3 : 2 : 1 (total 10).

Step 3: share = (ratio part / total of ratio) x cost. Rent to A = 15/40 x 10,000 = **3,750**. Repairs to A = 4/10 x 6,000 = **2,400**. Repeat for each department and each cost.

Step 4: total each column and check it equals 52,000. **Note** (professor): it is not stated whether fire insurance is for plant or stock; here it is taken on **stock**, so service dept D (no stock) gets nil.

Item	Basis	Amount	A	B	C	D
Rent	Area (15:11:9:5)	10,000	3,750	2,750	2,250	1,250
Repairs to plant	Plant value (4:3:2:1)	6,000	2,400	1,800	1,200	600
Depreciation of plant	Plant value (4:3:2:1)	4,500	1,800	1,350	900	450
Supervision	Wages (6:4:3:2)	15,000	6,000	4,000	3,000	2,000
Fire insurance	Value of stock (5:3:2)	5,000	2,500	1,500	1,000	-
Power	HP of machines (8:5:2)	9,000	4,800	3,000	1,200	-
Light	Light points (4:3:2:1)	1,000	400	300	200	100
Employer's insurance liability	No. of employees (4:3:2:3)	1,500	500	375	250	375
Total		**52,000**	**22,150**	**15,075**	**10,000**	**4,775**

Alternative: the rate method (slide: "Rate = OH / base"; the professor says *solve it both ways*). Rate for rent = 10,000 / 4,000 sq ft = Rs 2.50 per sq ft; A = 1,500 x 2.5 = 3,750; B = 1,100 x 2.5 = 2,750, and so on. Ratio method: Ratio = base / total base, then x OH. Rate method: Rate = OH / total base, then x each base. Same answer.

How to do it in Excel

Layout: base data block on top (rows 3-9, columns B to E = A, B, C, D), cost list below.

Cell	Content	Formula
A2:E2	Headers: Item, A, B, C, D	Type text
A3:E3	Area (sq ft)	1500, 1100, 900, 500
A4:E4	Plant value	240000, 180000, 120000, 60000
A5:E5	Wages, employees, HP, stock value, light points	Enter each as its own row
A12:A19	Cost items (rent, repairs...); amount in column F	e.g. F12 = 10000
Ratio method B12	Rent to dept A	=B$3/SUM($B$3:$E$3)*$F12
Rate method G12	Rate per sq ft	=F12/SUM($B$3:$E$3)
B12 (rate way)	Rent to A	=$G12*B$3
B20:E20	Total of each dept	=SUM(B12:B19)
F20	Check	=SUM(B20:E20)-SUM(F12:F19) (must be 0)

Steps: (1) type bases in rows; (2) type the first formula in B12 with **$ signs exactly as shown** so the base row (B$3) stays on the area row; (3) drag B12 right to E12; (4) drag the row down, but for each cost item change the base row number (e.g. rent uses row 3, repairs uses row 4); (5) for depreciation/repairs, copy the whole row of formulas and edit just the base row; (6) fill the supervision row with wages etc.
Tip: use blank cells for D under fire insurance and power (stock and HP are blank there), and the SUM will automatically give D nil.

Set 3, Problem 2: Direct redistribution method

Primary distribution summary: Production A 15,000; B 13,000; C 12,000 (total 40,000). Service: Stores 2,000; Time-keeping & accounts 1,500; Power 800; Canteen 500 (total 4,800). Overall 44,800. **Data (A : B : C):** HP 300 : 300 : 200; workers 20 : 15 : 15; value of stores requisitioned 2,500 : 1,500 : 1,000.

Professor's method

Rule: "this time there is no inter-service item; follow the same direct method": ignore services between service departments; split each service cost only among A, B, C.

Stores (value requisitioned 2,500 : 1,500 : 1,000 = 5 : 3 : 2): A = 5/10 x 2,000 = 1,000; B = 3/10 x 2,000 = 600; C = 2/10 x 2,000 = 400.

Time-keeping & accounts (workers 20 : 15 : 15 = 4 : 3 : 3): 600 : 450 : 450.

Canteen (workers 4 : 3 : 3): 200 : 150 : 150.

Power (HP 300 : 300 : 200 = 3 : 3 : 2): 300 : 300 : 200.

Costs	Basis	Total	A	B	C
As per primary distribution		40,000	15,000	13,000	12,000
Stores	Value requisitioned (5:3:2)	2,000	1,000	600	400
Time-keeping & accounts	No. of workers (4:3:3)	1,500	600	450	450
Canteen	No. of workers (4:3:3)	500	200	150	150
Power	HP of machines (3:3:2)	800	300	300	200
Total		**44,800**	**17,100**	**14,500**	**13,200**

Exam tip / watch out

The solution image in the Excel file prints Stores to B and C as "6000" and "4000": these are typing slips for **600** and **400**. The printed totals (17,100; 14,500; 13,200) are correct only with 600 and 400.

How to do it in Excel

Cell	Content	Formula
B3:D5	Bases (HP, workers, stores value) for A, B, C	Enter as rows
E8	Stores cost	2000
B8	Stores to A	=B$5/SUM($B$5:$D$5)*$E8
B9	Time-keeping to A	=B$4/SUM($B$4:$D$4)*$E9
B10	Canteen to A	=B$4/SUM($B$4:$D$4)*$E10
B11	Power to A	=B$3/SUM($B$3:$D$3)*$E11
B7	Primary total of A	15000
B12	Total of A	=SUM(B7:B11)
Drag	Copy B7:B12 to C and D	
E12	Check = 44,800	=SUM(B12:D12)

Steps: (1) put bases in rows 3 to 5; (2) put each service cost in column E; (3) write the formula once in B8, then drag right to D8 and down to B11:D11 after changing the base row; (4) total each production department; (5) verify E12 = 44,800.

Set 3, Problem 3: Step method

Departmental summary (primary distribution): Production X 12,000; Y 8,000 (20,000). Service: Stores 2,500; Time-keeping 2,000; Maintenance 1,500 (6,000). Total 26,000. **Data:** employees X 20, Y 15, Stores 10, Time-keeping 8, Maintenance 5; stores requisitions X 12, Y 10, Maintenance 3; machine hours X 1,200, Y 800.

Professor's method

Step 0: choose sequence. Time-keeping serves the most other service departments, then Stores, then Maintenance (the sequence given in the solution).

Basis: Time-keeping by employees: Stores 10 : Maintenance 5 : X 20 : Y 15 (total 50; its own 8 employees are excluded). Stores by requisitions: Maintenance 3 : X 12 : Y 10 (total 25). Maintenance by machine hours: X 12 : Y 8 (total 20).

Step 1: Time-keeping 2,000 -> Stores 400, Maintenance 200, X 800, Y 600. Time-keeping is now closed.

Step 2: Stores = 2,500 + 400 = 2,900 -> Maintenance 3/25 x 2,900 = 348; X 12/25 x 2,900 = 1,392; Y 10/25 x 2,900 = 1,160. Stores is now closed.

Step 3: Maintenance = 1,500 + 200 + 348 = 2,048 -> X 12/20 x 2,048 = 1,228.8 (about 1,229); Y 8/20 x 2,048 = 819.2 (about 819).

Step 4: X = 12,000 + 800 + 1,392 + 1,229 = **15,421**; Y = 8,000 + 600 + 1,160 + 819 = **10,579**; total **26,000**.

Dept	As per PD summary	1st step (Time-keeping)	2nd step (Stores)	3rd step (Maintenance)	Final
Time-keeping	2,000	(2,000)			-
Stores	2,500	400	(2,900)		-
Maintenance	1,500	200	348	(2,048)	-
X	12,000	800	1,392	1,229	**15,421**
Y	8,000	600	1,160	819	**10,579**
Total	**26,000**				**26,000**

How to do it in Excel

Cell	Content	Formula / value
B3:F3	Headers across: Time-keeping, Stores, Maintenance, X, Y	Type text
B4:F4	As per PD summary	2000, 2500, 1500, 12000, 8000
C6:F6	Time-keeping base: employees (Stores, Maint, X, Y)	10, 5, 20, 15
D7:F7	Stores base: requisitions (Maint, X, Y)	3, 12, 10
E8:F8	Maintenance base: machine hours (X, Y)	1200, 800
C10	1st step: Time-keeping share to Stores	=$B$4*C6/SUM($C$6:$F$6) (drag C10 to F10)
C11	Stores new total	=C4+C10 (= 2,900)
D12	2nd step: Stores share to Maintenance	=$C$11*D7/SUM($D$7:$F$7) (drag D12 to F12)
D13	Maintenance new total	=D4+D10+D12 (= 2,048)
E14	3rd step: Maintenance share to X	=$D$13*E8/SUM($E$8:$F$8) (drag to F14)
E16	Final X	=E4+E10+E12+E14 (= 15,421)
F16	Final Y	=F4+F10+F12+F14 (= 10,579)
G16	Check	=E16+F16 (must be 26,000)

Steps: (1) Put the departments across the top in the order of the steps (Time-keeping, Stores, Maintenance, X, Y). (2) Enter the three bases in their own rows; leave blank the cells for departments that are already closed or that do not use the service. (3) Write the first-step formula once in C10, using **$B$4** (locked cost) and **$C$6:$F$6** (locked total base), then drag across. (4) Add the share to the next department's own cost to get its new total (C11, D13). (5) Repeat the pattern for each step. (6) Final = own cost + all shares received; check the grand total is 26,000.

Set 3, Problem 4: Reciprocal services: repeated distribution and simultaneous equations

Given: Production A 8,000; B 6,500; C 7,000 (21,500). Service E 2,340; F 3,000 (5,340). Total 26,840.
Percentage charge-out: E: A 20%, B 25%, C 35%, F 20%. F: A 25%, B 25%, C 40%, E 10%. (Professor's margin note writes these as ratios: E -> A 4, B 5, C 7, F 4; F -> A 5, B 5, C 8, E 2.)

Professor's method

Method 1: Repeated distribution (cycles)

Put the service department totals under their columns, then repeatedly push the service totals out by the percentages until the leftover is too small to matter.

1. Cycle 1: E (2,340): A +468, B +585, C +819, F +468. F becomes 3,468.
2. Cycle 2: F (3,468): A +867, B +867, C +1,387.2, E +346.8.
3. Cycle 3: E (346.8): A +69.36, B +86.7, C +121.38, F +69.36.
4. Cycle 4: F (69.36): A +17.34, B +17.34, C +27.74, E +6.94.
5. Cycle 5 onwards: tiny balances (6.94, 1.39, 0.14...). Stop when negligible and allocate the remainder; totals converge to A 9,423; B 8,058; C 9,358.

Cycle	Distribute	A	B	C	Remaining service dept balance
Start		8,000	6,500	7,000	E 2,340; F 3,000
1	E 2,340	8,468.00	7,085.00	7,819.00	F 3,468.00
2	F 3,468	9,335.00	7,952.00	9,206.20	E 346.80
3	E 346.8	9,404.36	8,038.70	9,327.58	F 69.36
4	F 69.36	9,421.70	8,056.04	9,355.32	E 6.94
5	E 6.94	9,423.09	8,057.77	9,357.75	F 1.39
6	F 1.39	9,423.43	8,058.12	9,358.31	E 0.14
Final (rounded)		**9,423**	**8,058**	**9,358**	**total 26,839 (rounding)**

Professor's method (continued)

Method 2: Simultaneous equations (exact)

Let E and F be the **true total costs** of the two service departments (own cost + share of the other's cost):

- E = 2,340 + 0.10 F
- F = 3,000 + 0.20 E
- Substitute: E = 2,340 + 0.10 (3,000 + 0.20 E) = 2,340 + 300 + 0.02 E, so 0.98 E = 2,640, giving **E = 2,693.88**.
- Then F = 3,000 + 0.20 x 2,693.88 = **3,538.78**.

Now distribute the true costs to production departments:

	A	B	C	Total
Primary distribution	8,000.00	6,500.00	7,000.00	21,500.00
From E (2,693.88): 20% / 25% / 35%	538.78	673.47	942.86	2,155.10
From F (3,538.78): 25% / 25% / 40%	884.69	884.69	1,415.51	3,184.90

	A	B	C	Total
Total after reapportionment	**9,423.47**	**8,058.16**	**9,358.37**	**26,840.00**

Why reciprocal?

The step method only lets cost flow forward. In this problem E and F **use each other**, so a full two-way answer needs the reciprocal method. The repeated-distribution table is just a slow way of solving the same two equations; the equations give the exact answer in one go.

How to do it in Excel

Cell	Content	Formula
B2:B3	E and F primary costs	2340, 3000
B5:B6	% E gives to F; % F gives to E	0.20, 0.10
B8	E true cost (algebra)	=(B2+B6*B3)/(1-B5*B6) i.e. (2340 + 0.1 x 3000) / (1 - 0.2 x 0.1)
B9	F true cost	=B3+B5*B8
B11:D11	A, B, C primary	8000, 6500, 7000
B12	A total	=B11+0.20*$B$8+0.25*$B$9
C12	B total	=C11+0.25*$B$8+0.25*$B$9
D12	C total	=D11+0.35*$B$8+0.40*$B$9
E12	Check (26,840)	=SUM(B12:D12)

Option A (algebra): as in the table. **Option B (matrix, for many departments):** put the coefficient matrix **M** in G2:H3 = `1, -0.10 / -0.20, 1` and costs in J2:J3 = `2340 / 3000`. Select two cells and enter `=MMULT(MINVERSE(G2:H3), J2:J3)` (Ctrl+Shift+Enter in older Excel; just Enter in Excel 365). Result: 2,693.88 and 3,538.78. **Option C (repeated distribution):** build a table with one row per cycle; row 1: =B2 distributed; each next row uses the balance left by the previous row; copy rows down 6-8 times.

Practice problem: PlayCraft Inc. (toys): T-accounts, COGM, COGS, income statement

In simple words (layman view)

PlayCraft turns wood into toys. We follow the money through three buckets: **Raw materials -> Work in process -> Finished goods**, then onto the income statement. Two traps: buying a machine or office furniture is buying an **asset**, not an expense (only its depreciation counts), and recycled wood that cost nothing adds nothing.

Professor's method

Raw materials account: opening 75,000 + boards purchased 240,000 (direct) + supplies 60,000 (indirect) = 375,000 available; closing 45,000; **used = 330,000**, split into direct material 270,000 and indirect material 60,000. (DM consumed = 75,000 + 240,000 - 45,000 = 270,000, since all supplies were used.)

Factory overhead (MOH) 211,000: indirect material 60,000 + supervisor 50,000 + machine depreciation 36,000 + repairs 25,000 + factory electricity 40,000.

WIP account: opening 20,000 + DM 270,000 + DL 150,000 + MOH 211,000 = 651,000; closing 30,000; **COGM = 621,000**. **Total manufacturing costs = 270,000 + 150,000 + 211,000 = 631,000.**

FG account: opening 60,000 + COGM 621,000 = 681,000 available; closing 70,000; **COGS = 611,000**.

Not product costs: office depreciation 8,000, administrative salaries 20,000, advertising 35,000 and freight outward 45,000 are period costs. The machine purchase (4,00,000) and furniture (80,000) are balance-sheet assets.

Raw materials inventory	Rs	Work-in-process inventory	Rs
Opening	75,000	Opening	20,000

Raw materials inventory	Rs	Work-in-process inventory	Rs
DM purchased	240,000	+ DM used	270,000
Indirect material purchased	60,000	+ DL	150,000
		+ Manufacturing overhead	211,000
Available	**375,000**	**WIP AFU**	**651,000**
- DM used	(270,000)	- Closing WIP	(30,000)
- Indirect material used	(60,000)	**= COGM (to FG)**	**621,000**
Closing RM	**45,000**		

Finished goods inventory	Rs
Opening	60,000
+ COGM	621,000
FG AFS	**681,000**
- Closing FG	(70,000)
COGS	**611,000**

(b) Required amounts	Rs
Direct material consumed	270,000
Total manufacturing costs (DM + DL + MOH)	631,000
Cost of goods manufactured	621,000
Cost of goods sold	611,000

(c) Traditional income statement	Rs
Sales revenue	1,200,000
Less: COGS	(611,000)
Gross profit	**589,000**
Operating expenses: freight outward	(45,000)
Depreciation (office furniture)	(8,000)
Administrative salaries	(20,000)
Advertising	(35,000)
Operating profit	**481,000**

Exam tip / watch out

The professor's solution sheet prints "Direct material consumed 200,000" in part (b), which is a slip. Its own T-accounts, and the arithmetic above, give **270,000** (and total manufacturing costs 631,000 only works with 270,000).

Answer to (d)

(d) Which costs rise with production? Variable costs rise with output: direct materials, direct labour (if paid per piece), supplies/indirect materials, and often factory electricity. **Fixed** costs stay unchanged in the short run if present capacity is enough: machine depreciation (36,000), factory supervisor salary (50,000), and most repairs contracts. **Why it matters:** management can estimate the *extra* cost of producing more (mostly variable costs), see that cost *per unit* falls as fixed costs are spread over more units, and use this for production plans and pricing. Also remember unit costs must always be tied to a given output level.

How to do it in Excel

Cell	Content	Formula
B2:B4	Opening RM, DM purchased, indirect material	75000, 240000, 60000
B5	Closing RM	45000
B6	DM consumed	=B2+B3-B5
B7	Indirect material consumed	=B4
B9	MOH	=B7+50000+36000+25000+40000
B10	Total manufacturing costs	=B6+150000+B9
B11	COGM	=20000+B10-30000
B12	COGS	=60000+B11-70000
B14	Gross profit	=1200000-B12
B15	Operating profit	=B14-45000-8000-20000-35000

Steps: (1) first **list every transaction and tag it**: DM / indirect material / DL / MOH / period cost / asset (ignore). (2) Put the opening balances in cells. (3) Build the three T-accounts side by side: debit side left, credit side right, and use `=SUM()` to total each side; the closing balance is the number that makes both sides equal (`=total debit - other credits`).

PART F: Quick Revision Sheet

One-page formula sheet

Item	Formula
DM AFU	Opening DM + purchases + freight - returns
DM used	DM AFU - closing DM
Prime cost	DM used + DL (+ direct expenses)
Conversion cost	DL + MOH
TMC = COP gross = works cost gross	Prime cost + factory overhead
WIP AFU	TMC + opening WIP
COGM = works cost net	WIP AFU - closing WIP
FG AFS	COGM + opening FG
COGS	FG AFS - closing FG
Gross margin	Sales - COGS
Operating profit	Gross margin - (selling + admin expenses)
Total cost (cost sheet)	COGS + admin OH + selling & distribution OH
Cost per unit	Total cost / units
Fixed cost per unit	Total fixed cost / units (falls as units rise)
Differential cost	Cost(alternative 1) - Cost(alternative 2)
Profit x% on cost	Price = cost x (1 + x)
Profit y% on selling price	Price = cost / (1 - y)
Share by ratio	(part / total of ratio) x cost
Rate method	Rate = OH / total base; share = rate x base
Reciprocal (2 service depts)	E = own_E + %F-to-E x F; F = own_F + %E-to-F x E

Common mistakes to avoid

- Putting purchase returns on the wrong side: returns **reduce** purchases.
- Forgetting **accrued** wages/expenses: add them to the paid amounts.
- Including interest, bad debts, dividend income in cost accounts: they are financial items.
- Treating a machine purchase or furniture as an expense: only **depreciation** is a cost.
- Treating selling/admin costs as product costs: they are **period costs** (unless your problem's cost sheet adds admin into cost of production).
- Multiplying a fixed cost per unit by a new volume: fixed costs are fixed **in total**.
- Using a unit cost without stating the output level.
- In step method, charging cost back to a department that has already been closed.
- Not checking that distributed totals equal the original totals.

Glossary (A to Z)

Term	Plain meaning
Absorption	Charging overheads to products using a rate
AFU / AFS	Available for use / Available for sale
Allocation	Charging a whole identifiable cost to one department/product
Apportionment	Sharing a cost between departments by benefit
Budget	Numerical plan; main planning tool
Committed fixed cost	Long-term fixed cost you cannot cut quickly
Conversion cost	Cost of turning materials into product: DL + MOH
Cost object	Anything we want the cost of

Term	Plain meaning
COGM	Cost of goods manufactured (completed in the period)
COGS	Cost of goods sold (period expense)
Differential cost	Difference in cost between two choices
Direct cost	Traceable to the cost object
Discretionary fixed cost	Fixed cost managers can change short-term (advertising)
Indirect cost	Not traceable; allocated
Marginal cost	Cost of one more unit
MOH	Manufacturing overhead: all factory costs except DM and DL
NMC	Non-manufacturing (period) costs: selling and admin
Opportunity cost	Benefit of the best alternative given up
Period cost	Expensed in the period incurred
Prime cost	DM + DL
Product cost	DM + DL + MOH; held in inventory until sold
Relevant cost	Future cost that differs among alternatives
Relevant range	Activity band where cost behaviour assumptions hold
Sunk cost	Already spent; irrelevant for decisions
TMC	Total manufacturing cost = COP gross
WIP	Work in process: started, not finished